DFT methodologies for testing k-CNOT, Fredkin and Peres based Reversible Circuits

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***Abstract*—The widespread use of reversible logic for low-power and high-speed applications has made testing of reversible circuits a primary focus for researchers in order to ensure the manufacture of highly reliable circuits. In this work, we propose two methods to convert a given Reversible Circuit (RC) into a testable RC. Initially, we present the modification to ESOP (EXOR Sum of Product) based online testing methodology with additional test vectors to detect the wide class of faults in RC designed with k-CNOT, Fredkin and Peres gates.Then, we propose a new Design for Testability (DFT) technique which effectively detects various faults, using a minimum test set for higher fault coverage with additional DFT hardware. The proposed methodologies are validated on various reversible benchmark circuits and comparison of fault coverage and hardware overhead are compared with that of existing approaches.**

***Index Terms*—DFT, Online testing, Reversible circuits, ATPG**

I. Introduction

The association of logical irreversibility with physical irreversibility was discussed in [1]. The standardization of signals for minimizing heat per cycle was explained and also defined that if the output of the device did not uniquely define the inputs, then it is said to be logically irreversible. The energy loss for every bit loss 'E' is less than energy limit proposed by [1], $KT \times ln2$ [K= Boltzmann constant, T= Absolute temperature]. Therefore, reversible computing has become a low-cost alternative to conventional computing in terms of power consumption, speed and computing capability which is of interest among the researchers in quantum field [2].

A fault is a physical disturbance which alters the functionality of the circuit. For any input test vector, if the outputs of the faulty and the normal circuit are different, the fault is said to be detectable. Similarly, a fault is redundant when the circuit functionality does not change irrespective of any test input. In both reversible logic and conventional circuits, for all the fault models, the complexity of generating test vectors can be shortened through the use of those models that comprise variable fault possibilities. The total number of fault sites is defined as the summation of input and output fault sites for every level in the circuit. Stuck-at-Faults (SAF) occur if the signal at a fault site is set to either logic 0 or 1. SSF (Single Stuck-at-Fault) and MSF(Multiple Stuck-at-Fault) are two catagories of SAF. Details about these faults are in [3] and [4]. Cross-point faults(CPF) refers to faults where controls in the reversible gates go missing [5]. Different forms of missing gate faults(MGF) are single MGF, multiple MGF, partial MGF where all or few control points of a gate are missing or not operative and repeated gate fault(RGF) with redundant gates several times erroneously. A defect where lines inadvertently touch each other leading to shorts is referred to as Bridging Faults. If the two input or output lines are shorted to logic 0, then it is called AND bridging whereas if they are shorted to logic 1, then it is called OR bridging. Bridging faults can be also categorized as intra-level bridging as well as inter-level bridging. Reversible circuit have unique one to one mapping between their input and output signals which makes it distinct from conventional circuits. The various fault models for Reversible Circuits can be found in [6] and [7].

Testing is an essential phase in design to guarantee availability, reliability, and quality of a device or circuit. The two types of testing are offline or non-concurrent testing and online or concurrent testing. In offline testing, different test patterns are required for each fault at every fault site. The circuit is required to be in test mode where it is withdrawn from its normal functionality. Here, a time overhead exists. In online testing or DFT, the circuit can be tested in its normal operation with some additional test hardware [8]. Authors in [9] have come up with an approach which has been extended for the online testing of RC. They have proposed an online testable substitution for $n \times n$ reversible logic gate. In the research by authors in [10], a design methodology for online testing of Toffoli network was proposed which was tested for three fault models (MGF, CPF and single bit faults) by converting every Toffoli gate into parity preserving gate by adding 1 parity line and 2 CNOT gates. In the research work [8], authors have discussed the use of a Boolean function for detecting an online fault in the reversible logic circuit for all types of Toffoli networks. In [11], two categories of reversible circuits i.e. information reversible circuits and energy reversible circuits was presented. SAF, BF, MGF and Delay faults come under information RC whereas SAF, Missing control faults(MCF), BF and quantum faults come under energy RC. In this paper, the faults under both these categories have been considered.
The comprehensive and comparative study for online testability of reversible circuits by authors in paper [12] focus on SAF and MGF using additional hardware overhead. This

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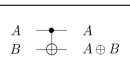
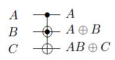
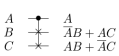
work initially presents the modification to the existing online ESOP proposed in [3] which is referred to as modified ESOP method. The paper also presents a new DFT methodology for detecting various faults. Various fault models are tested using both approaches for RCs designed with k-CNOT, Peres and Fredkin gates. The paper is organized as follows: section II describing the preliminary work on fault models and DFT based testing of RC. In section III, modified ESOP and the proposed DFT methods are discussed. Section IV presents the results and comparison with existing work which is followed by the Conclusion and the References.

II. Preliminaries on fault detection in RC

Feynman, Toffoli, Peres and Fredkin are the standard reversible gates. Their symbol, functionality and the cost related to their function are tabulated in Table I. In the proposed work, the testing methodologies developed are validated on circuits designed with these standard gates.

TABLE I: Reversible Gates

Gate	Circuit	Quantum Cost and Delay
Feynman		1,1Δ
Peres		4,4Δ
Fredkin		5,5Δ

A. *Existing work on online and DFT approaches for reversible circuits*

The comprehensive study on the review of online testable approaches is presented in [12] for suitable fault model [12]. In [13], the author proposed new DFT method for single bit fault in RC designed with k-CNOT gates. In [14], the Online testability using R1 and R2 gates for single stuck-at fault in k-CNOT based RC is proposed. This method used parity as a measure for fault detection.In [15], the Online testability for single stuck-at faults using TRC (testable reversible circuits)method, any $x * x$ gate is converted to $(x+1)*(x+1)$ reversible gate the extra input and output being that of parity. This method also considered the single bit fault in RC designed with k-CNOT gates. Authors in [16] proposed a method based on Dual Rail Coding approach. Although for this method the percentage of fault coverage is 100% it has an overhead of extra lines for every input and output line. The author in [17] presented online DFT technique for detection of bit faults and bridging faults in k-CNOT based RC. They claim that only $log_2 N + 3$ vectors are sufficient to test any given RC with additional hardware based on the methodology presented. Work in [18] has discussed on conversion of the various reversible gate to online testable and also discussed on detection of single and multiple bit faults k-CNOT based RC. Online detection of 3 fault models is presented in [10]. The authors in [10] has analyzed detection of faults using the methodology developed. The hardware complexity of their method was 2(L+N) where L is the number of gates and N is the number of lines. In [13], the author has proposed a new DFT method for detecting bit faults in k-CNOT based RC. The same authors [13] have presented a survey of existing online testing methods along with the type of fault detected using existing methods [12]. A test methodology for detecting SMGF is proposed in [19] which considers reversible circuits designed with positive and negative controlled k-CNOT gates. There are various offline testing methods found in [4], [20]–[25]. These offline techniques detect faults without any hardware overhead but the circuit needs to be configured to work in Test mode.

B. *Motivational example*

The existing ESOP technique is suitable for bit faults and it can detect any single or multiple bit faults. If the same technique is applied to other fault models, the technique cannot detect all faults. According to the existing ESOP based method for online testing for k-CNOT based RC in [3], with any normal input test vector applied, any fault in the circuit is said to be detected. According to the authors in [10], the faults in Preamble block goes undetected since the fault detection is based on parity check. Hence the need for a generic online testing methodology is required targeting a wide class of fault models and gate libraries. The illustration shows the undetected fault condition in an example circuit using the approach in [3] for stuck-at fault model.

Stuck at fault model: Consider the stuck at 1 fault induced at the output of 1st input line in the first gate for illustration. According to ESOP, any test vector can be chosen to detect the fault. Hence, considering a random test vector 1100 as shown in Fig 1, the fault at the output of 1st level is not activated since it must be forced to opposite value being logic 0. Even the parity line fails to drive L to opposite value i.e logic 1. In this

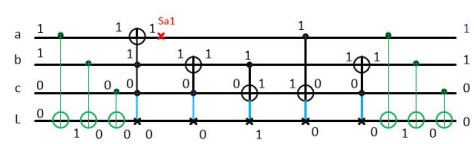


Fig. 1: Stuck at 1 fault induced at 1st level at output of 1st input line

work, we propose a modification to the existing method in [3] along with a test vector generation methodology to maximize the fault coverage. To enhance the fault coverage for a set of all missing gate faults, a new DFT method is also proposed. The proposed methods are validated for various fault models on an RC designed with k-CNOT, Peres and Fredkin gates.

III. Proposed Testing methodologies

To detect various faults using ESOP technique of [3], we propose modifications to existing ESOP in terms of the test application. Retaining the existing hardware structure, by exercising with test vectors, a large number of faults can be detected.

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A. *Proposed modified ESOP method*

In the modified ESOP method, a certain number of test vectors are chosen to activate the fault and propagate it to obtain high fault coverage. This method has given very good results for stuck-at faults and bridging faults. But the drawback is in the case of the missing gate or missing control fault as well as additional control fault where for given a certain number of test vectors, 100% fault coverage is not achieved.

Algorithm 1 Modified ESOP Method

INPUT$(RC, InputFaultType)$
OUTPUT$(TestSet, AdditionalHardware, FaultCoverage)$
use(RC)**get**(N(Input lines),G(Gates)) ▷ Input parameter extraction
use(RC)**get**(C(Control lines),T(Target lines)) ▷ N=C+T
procedure SELECTION(C, T) ▷ To generate test vectors
  for $i \leftarrow 1, G$ **do**
    Extend gates to preserve parity with additional line 'L' initialized to 0 ▷ As in [3]
  end for
  for $i \leftarrow 1, N$ $with$ $G = 1$ **do**
    Set $C = 1, T = 1 \oplus C$
      Pad all zeros or all ones to the other inputs in the test vector and propagate fault at 'i'
    Set $C = 0, T = 1 \oplus C$
      Pad all zeros or all ones to the other inputs and propagate fault at 'i'
    Set all zeros or all ones to the inputs of test vector and propagate fault at 'i'
  end for
  $AdditionalHardware \leftarrow ((2 \times N) \times CNOT) + (G \times CNOT)$
  $TestSet \leftarrow Minimum\ required\ set\ of\ vectors$
end procedure

Illustration for SAF: The selection of test vectors for various fault models with modified ESOP is explained with the help of an example. A benchmark circuit $ham3$ with Swap gate implementation is taken from [26] is considered. The number of gates G = 4 and input lines N = 3. $Additionalhardware = 2 \times 3 + 3 = 9$ gates. The stuck-at 0 fault for the first gate and first input line is considered for illustration. Referring to the Algorithm 1, the fault is activated by forcing it to opposite value i.e., logic 1 at the particular line 'j' and it is propagated throughout the circuit. The parity line after traversing 'G' gates goes high indicating fault detection. Hence test vectors for fault detection are 1000,1100,1110.
Feasibility of test vector: If we consider 1100 or 1000 as test vectors, the output at the first gate would be 0100 or 0000 respectively. So, the stuck-at 0 fault cannot be propagated. Hence, the correct test vector to be chosen is 1110 for fault propagation.

Missing Gate fault model: Let us consider the second gate

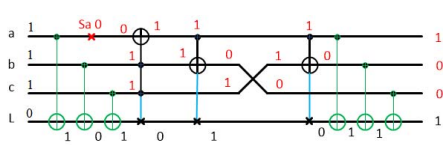

Fig. 2: Stuck at 0 fault induced at 1st level at 1st input line

is missing in Fig.3. The resulting circuit with fault induced is shown in Fig.3. To detect missing gate fault, all the control lines C are forced to 1 and remaining target lines to be 0.So the test vector suitable for this fault is 1110. We can observe that, though the fault is propagated and faulty and fault-free outputs are different, the parity line could not detect the fault.Hence, using modified ESOP method, there are some faults which go undetected by parity line. **Bridging fault model**: In the

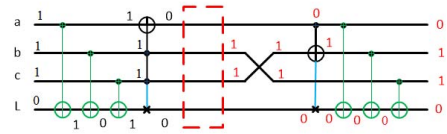

Fig. 3: Missing Gate fault induced at 2nd level

case of bridging fault, the fault activation requires the input lines which are bridged, to be forced to opposite values. The modified ESOP method generates the test vector which can detect all bridging faults. The detailed analysis is shown below. Here an illustration of AND bridging at the 4th gate for the first and second input lines is given. The test vector selection for bridging fault is that the value of inputs before the bridging must be 01 or 10 and after bridging it'll be ANDed to 00. Hence the test vectors suitable are 1100,1110. For this model, the fault detection is 100% using modified ESOP method.

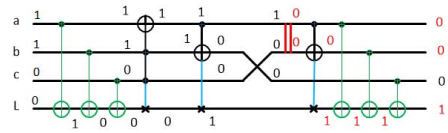

Fig. 4: AND Bridging fault induced at 4th level between 1st and 2nd line

B. *Proposed DFT Methodology*

In any reversible circuit, the target of a gate captures the control signal values and the target line input. Based on control line values, the target changes/flips. Thus tracking the target at every level of the circuit can be used for fault detection. So far in the literature, there are various works on online testing of RC. These online techniques focus on parity preserving technique and are suitable to detect bit faults or stuck at faults. The main idea in parity preserving gate helps detect any cell faults, bit flip faults, since they may impact the output line directly. The same technique cannot be extended for missing gate faults since this fault may or may not change the parity as the effect of these faults can be observed in target lines. Some techniques in literature propose new testable gates based on parity preserving. In this work, we try to develop an online DFT method for testing k-CNOT, Fredkin, and Peres based RC for various fault models. In the new DFT method, instead of CNOT gates at input and output, the CNOT gate is added only at the output of the target line at every gate (which we referred as a level) and every gate function is extended to DFT line L.

C. *Fault detection methodology used in proposed DFT methodology*

The DFT line L will capture all the signals on control and target of any gate. For illustration, if we consider Fig.6, output of the DFT line at various levels is

- After level 1 $L_1' = L \oplus C1C2$
- After level 1 and insertion of CNOT to target of level 1 gate $L_1'' = L \oplus C1C2 \oplus T1$
- After level 2 $L_2' = L \oplus C1C2 \oplus T1 \oplus C_2'$
- After level 2 and insertion of CNOT to target of level 1 gate $L_2'' = L \oplus C1C2 \oplus T1 \oplus C_2' \oplus C_1'$
- After level 3 $L_3' = L \oplus C1C2 \oplus T1 \oplus C_2' \oplus C_1' \oplus C_1'$ = $L \oplus C1C2 \oplus T1 \oplus C_2''$ (since $C_1' = C_1''$)
- After level 3 and insertion of CNOT to target of level 1 gate $L_3''' = L \oplus C1C2 \oplus T1 \oplus C_2'' \oplus C_2'''$

Here, T is the target line, C1 and C2 are initial controls, C_1' and C_2' are the controls after Level 1 and so on. To explain the fault detection methodology of our proposed DFT, we take a simple RC as shown in Fig.6. Initially, all the gate functions are extended to DFT line L and from every target of a gate, a CNOT gate is added to DFT line. Since the example circuit contains 3 gates, it requires 3 gates for extending the gate and 3 additional CNOT gates for extending the actual function of the gate to DFT line. The actual function extracted as DFT line L is $L \oplus a \oplus c \oplus b$.

The stuck-at-0 fault is induced at second input 'b' in level 1, the functionality at the DFT line L changes and the faulty variable which is represented as $\bar{b}$ will change the function propagated to the output of DFT line. The DFT line L output for the faulty circuit is obtained as $L \oplus a \oplus c \oplus b$ as shown in Fig.6. Thus the output obtained in DFT line will be mutually exclusive for fault-free and faulty situations in the circuit and thus indicates the fault. The same analysis when extended for Missing gate faults, the faulty and fault-free outputs are mutually exclusive.For example in Fig.5, if gate 2 is missing the output at DFT line L will change to $L \oplus a \oplus b$ which again captures the fault. The technique fails when multiple stuck-at faults in situations where the fault propagation will not make the faulty and fault-free behavior of the circuit different. For an example, if simultaneously, a stuck-at-1 fault in line a and stuck-at-zero fault in line b will over right the DFT output and both faulty and fault free outputs are the same masking the fault effect. Thus the proposed method is well suited for single stuck-at-faults, missing gate faults and bridging faults.

Illustration

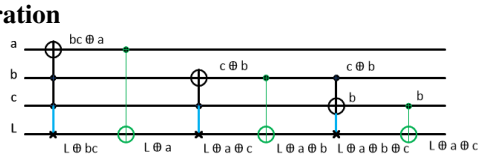

Fig. 5: Functional description of fault free partial ham3 circuit

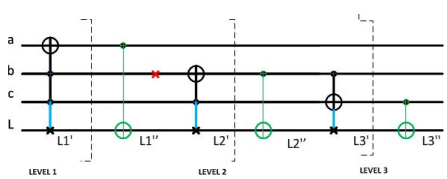

Fig. 6: Functional description of stuck-at-0 fault induced at 2nd input

The selection of test vectors for different fault models under new DFT method is given in Algorithm 2. For the illustration of different fault models, $ham3_tc$ circuit is considered. The number of gates G = 5 and input lines N = 3. Number of control lines C = 6 and number of target lines T = 5. Additional

Algorithm 2 Proposed DFT Methodology

INPUT$(RC, InputFaultType)$
OUTPUT$(TestSet, AdditionalHardware, FaultCoverage)$
use(RC)**get**(N(Input lines),G(Gates)) ▷ Input parameter extraction
use(RC)**get**(C(Control lines),T(Target lines)) ▷ N=C+T
procedure SELECTION(C, T) ▷ To generate test vectors
  for $i \leftarrow 1, G$ **do**
    Extend gate function to the additional DFT line 'L' and initialize 'L' to 0
  end for
  for $i \leftarrow 1, N$ $with$ $G = 1$ **do**
    Set $C = 1, T = 1 \oplus C$
      Pad all zeros or all ones to the other inputs in the test vector and propagate fault at 'i'
    Set $C = 0, T = 1 \oplus C$
      Pad all zeros or all ones to the other inputs and propagate fault at 'i'
    Set all zeros or all ones to the inputs of test vector and propagate fault at 'i'
  end for
  $AdditionalHardware \leftarrow (T + G) \times CNOTgates$
  $TestSet \leftarrow minimum\ test\ vector\ required$
end procedure

hardware = 5+5 = 10 CNOT gates. The difference for each fault lies in the test vector generation explained below.

Stuck at fault model: Stuck-at-0 fault at second gate and second input line is considered for illustration. Referring to Algorithm-2, the fault can be activated by passing the opposite value 1 at the particular line 'j'(stuck at 1 fault is activated by passing the opposite value 0). The subset of test vectors that can be suitable for detecting the fault would be 0110,1100,1110. **Missing Gate fault model**: For an

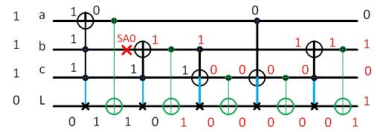

Fig. 7: Stuck at 0 fault induced at 2nd level in ham3_tc circuit

illustration of missing gate fault, the third gate is considered as missing. According to Algorithm 2, the test vectors are chosen such that all the control inputs are at logic 1 and parity line input is chosen to be 0. Hence the test vector to be used to detect any missing gate fault is 1110.

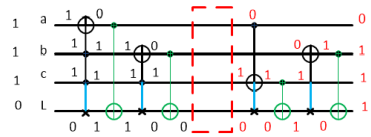

Fig. 8: Missing gate fault induced at 3rd level

Additional Control fault model: Referring to Algorithm 2 mentioned in the previous section, in order to choose the test vector for this fault model, the additional control bit must be set to logic 0 and the other default control bit is to be made logic 1. In the illustration below for the fifth gate which is functionally represented as $ac \oplus b$, the control input c is chosen to be 1 and additional fault at 'a' is made 0, so that $ac \oplus b$ becomes $0 \oplus b$ due to 'a'. Hence, the test vector is chosen to be 1110 in the fifth level of the ham3 circuit.

Bridging Fault model: The illustration shows OR bridging for

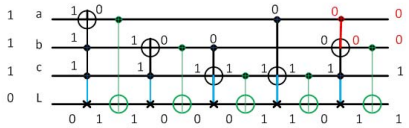

Fig. 9: Additional control fault induced at 5th level for ham3 circuit

the second gate. The fault activation requires the input lines which are bridged to be forced to opposite values. And the output lines after bridging are forced to logic 1 in the case of OR bridging. Accordingly, the test vectors that can be chosen are 1100,0110.

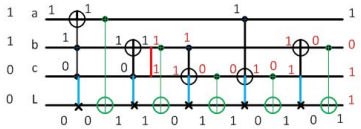

Fig. 10: OR Bridging introduced at fourth level between 2nd and 3rd output lines

A similar analysis can be extended for MCF and AND BF models.

IV. Results and Discussions

The proposed test algorithms are applied on various benchmark circuits [26]. Table II gives the comparison of test vector required and fault coverage for $ham3_tc$ circuit for all fault models. It is evident that the modified ESOP based method provides 100% fault coverage in case of stuck-at and bridging faults. This maximized fault coverage is due to the fact that the proposed modified ESOP method uses a test vector set. For the set of all missing gate faults, the proposed DFT method gives 100% fault coverage indicating the superiority of the proposed DFT method over the modified ESOP.

TABLE II: Comparison of Modified ESOP and DFT methods for ham3 circuit

FAULT MODEL	METHOD	QC	TEST VECTOR	FAULT COVERAGE
Stuck at 1 Fault	ESOP	20	0000,0110	100%
	DFT	19	0000,1110,1000,0010	50%
Stuck at 0 Fault	ESOP	20	1110,1100	100%
	DFT	19	1110,1100,1010	54.5%
Missing Control Fault	ESOP	20	1110,0100,1100	100%
	DFT	19	1110,1100,1000	80%
Additional Control Fault	ESOP	20	1110,0100,1000	100%
	DFT	19	1110,1000	75%
Missing Gate	ESOP	20	1110	20%
	DFT	19	1110	100%
Bridging Fault (AND)	ESOP	20	1100,0110,1110	100%
	DFT	19	1100,0110,1000	85.7%
Bridging Fault (OR)	ESOP	20	1000,1100,0110	100%
	DFT	19	1100,0110,1000	50%

A similar observation is evident from the result tabulated in Table. III and Table. IV for other benchmark circuits. In Table. III, for nth_prime circuit, the proposed DFT method performs better to detect missing gate and missing control fault when compared to modified ESOP method. Although, a 100% coverage is obtained for stuck-at-fault and bridging faults using the modified ESOP method.

For $ham3_tc$ circuit using swap gate, exhaustive testing was done for different fault models and following is the Table IV that shows the analysis results. New DFT method can be used

TABLE III: Comparison of Modified ESOP and DFT methods for nth prime circuit

FAULT MODEL	METHOD	QC	TEST VECTOR	FAULT COVERAGE
Struck at Fault 1	ESOP	18	0000,0011,1000,1100	100%
Struck at Fault 0	DFT	18	1110,1100,0110	100%
Missing Control Fault	ESOP	18	1110,1010	75%
	DFT	16	1110.1010	75%
Additional Control Fault	ESOP	18	0110,1000	100%
	DFT	16	0110,1000	67%
Missing Gate	ESOP	18	1110,1100	50%
	DFT	16	1110,1100	100%
Bridging Fault (AND)	ESOP	18	1100,0110,1000	100%
	DFT	16	1110,1100,1000,0110	91.66%
Bridging Fault (OR)	ESOP	18	1100,0110,1000	100%
	DFT	16	1110,0110,0011,1000,1100	75%

as a generalized method for all fault models but with the shortcoming of stuck-at and bridging fault, which does not give 100% fault coverage due to redundant fault sites or fault propagation. For missing gate and additional control faults, the DFT method proposed gives 100% results with just two test vectors whereas a set of two vectors is insufficient to detect these faults in the case of modified ESOP method.

TABLE IV: Comparison of Modified ESOP and DFT methods for ham3 (using swap gate) circuit

FAULT MODEL	METHOD	QC	TEST VECTOR	FAULT COVERAGE
Struck at Fault 1	ESOP	18	1100,0110,1000	100%
Struck at Fault 0	DFT	18	1110,1100,0110	100%
Missing Control Fault	ESOP	18	1110,0110	100%
	DFT	15	1110,0110	100%
Additional Control Fault	ESOP	18	1100,0110	0%
	DFT	15	1100,0010	100%
Missing Gate	ESOP	18	1110,1000	0%
	DFT	15	1110,1000	100%
Bridging Fault (AND)	ESOP	18	1000,1100,0110	100%
	DFT	15	1100,1000,1110,0110	100%
Bridging Fault (OR)	ESOP	18	1000,1100,0110	100%
	DFT	15	1100,1000	91.66%

Similarly, the analysis is performed for $rd32$ benchmark circuit and results are shown in Table. V. Here, using new DFT method with a single test vector resulted in a 100% fault coverage for Missing gate faults which surpasses the modified ESOP method. It is evident from the analysis and tabulations that the fault coverage with new DFT method for missing and additional faults are either equal to or greater than that of modified ESOP.

TABLE V: Comparison of Modified ESOP and DFT methods for rd32 circuit

FAULT MODEL	METHOD	QC	TEST VECTOR	FAULT COVERAGE
Struck at Fault 1	ESOP	24	11100,11000,01100	100%
	DFT	20	10000,01100,11100	100%
Missing Control Fault	ESOP	24	11100,01100	75%
	DFT	20	11100,01100	75%
Additional Control Fault	ESOP	24	11000,10000,01100	100%
	DFT	20	11000,10000,01100,10000	67%
Missing Gate	ESOP	24	11100	0%
	DFT	20	11100	100%
Bridging Fault (AND)	ESOP	24	10000,11100,01000,11000	100%
	DFT	20	01100,11100,11000	87.5%
Bridging Fault (OR)	ESOP	24	10000,11100,01000,11000	100%
	DFT	20	01100,11000,11100	82%

For a given RC, the hardware complexity for the proposed DFT is Gate overhead = (Number of gates) + (Number of targets) = G + T and Quantum cost overhead = QC of original circuit + total number of targets * CNOT gates= QC+T (since QC of CNOT is 1). Consider n as number of lines and G as number of gates and T as number of targets.

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TABLE VI: Hardware overhead in terms of Quantum cost compared to [8]

bench mark circuit	work in [8]			proposed work DFT			QC overhead with [8]
	n	N	QC	n	additional QC	actual QC	
rd32	5	9	45	6	67	29	48.89
majority	7	36	154	7	308	136	100
xor5	5	4	22	6	12	4	-45.4
alu(mini-alu)	20	58	82	11	178	60	117.07
ham3	4	21	25	4	35	7	40
mod5	6	25	29	6	49	12	68.97
nth_prime3	4	18	22	4	30	6	36.36
permanent_2x2	7	21	54	7	99	39	83.33

TABLE VII: Hardware overhead of proposed DFT methodology for various benchmark circuits

benchmark circuit	n	GC	QC	QC after DFT	GC after DFT	% QC overhead	% GC overhead
ham3tc	3	5	7	19	10	171	100
graycode6	6	5	5	15	10	200	100
hwb4-11-23	4	11	23	57	22	147.	100
4_49_12_32	4	12	32	76	24	137.5	100
mod5adder_15	6	15	83	181	30	118	100
ham7tc	7	23	81	185	46	128	100
cycle17_3	20	48	6057	12162	96	100	100
mod1024adder	20	55	1541	3137	110	103	100
ham15tc1	15	132	1827	3786	264	107	100
hwb9_1544	9	1544	44702	90948	3088	103	100
hwb10-3631	10	3631	139470	282571	7262	102.6	100
hwb11-9314	11	9314	345020	699354	18628	102.6	100

V. Conclusion and Future Work

In this work, we have proposed a new DFT technique for testing of k-CNOT, Fredkin, and Peres based reversible circuits which can test a circuit for bit faults, set of all missing gate faults and bridging faults. The fault models considered in this work are appropriate for Information reversible and energy reversible circuits. Extensive study of proposed modified ESOP and new DFT method on various benchmark circuits reveals that modified ESOP works very well with bit faults and bridging faults. The new DFT works efficiently for detecting missing gate faults. With a minimum number of test vectors, higher fault coverage is achieved for SSF, MSF, MGF, MCF and BF fault models. The proposed DFT method can be used to test RC designed with k-CNOT, Fredkin, and Peres gates by capturing the functionality in DFT line L unlike the parity check methods reported in the literature which requires the conversion of any gate into a parity preserving gate. The hardware overhead of proposed new DFT method is proportional to the number of gates. The quantum cost overhead is 2(original QC of the circuit)+ T and the Gate count is 2(original GC of the circuit). Since the proposed method has high fault coverage with large fault set, the proposed method can be adopted for testing quantum reversible circuits to gain confidence over the implemented designs. Our future work is to investigate the online testing methodology targeting less hardware overhead, enhanced fault coverage and devise an algorithm for new fault models.

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